When training models in a notebook, initiating an MLflow run with `mlflow.start_run()` is the recommended practice. This function begins a new experiment run. Any subsequent training information, such as parameters, metrics, and models, is automatically tracked and logged by MLflow Tracking until the run ends. In *Applying our learning* for the Streaming Transactions project, we use autolog as well as add additional parameters to be tracked. To customize the autologging configuration, call `mlflow.autolog()` before your training code. Please note that although many common libraries have autolog support (such as Scikit-learn, XGBoost, and Keras), not all do. Check the documentation linked in *Further Reading* for the latest list.

ML Artifacts

Artifacts in the context of experiment tracking are any physical files, images, or models generated while training a machine learning model. These are stored and logged for future reference and analysis.

Grouping these runs, or executions of modeling code, into experiments when working on a particular ML task makes it easier to compare evaluation metrics and other details. In the next section, we invite you to explore options to leverage external data and models.

Integrating external data and models

The AI and ML world moves extremely fast. You will likely come across projects that benefit from innovations and data outside your DI Platform. For example, it's common to augment corporate datasets with external data. Similarly, it is also common to augment an ML solution with a pre-trained model. Your platform should never be a blocker to open-source options, which is why Databricks offers several paths to integrate external data resources into your projects. To demonstrate, we have included examples of three different methods to combine external data and models with your own data artifacts: downloading data from the **Databricks Marketplace**, accessing OpenAI's ChatGPT model via API, and exploring the **LLM Playground**. Please note that the following examples are outside of the *Applying our learning* project work, so while we encourage you to follow along, the Marketplace and LLM Playground work will not impact the example projects themselves.